COAL QUIZ 3 (B)

NAME: ____________________________ SECTION______ STUDENT ID: ____________

1. Update flags after arithmetic instruction. Also state which of the following jumps will taken or not taken

```
.code
   mov ax,07B1Ah
   sub ax,0CEEBh
    jc l1
L1: jz L2
L2: jo L3
L3: js L4
L4: jp L5
L5:
```

	TAKEN	NOT TAKEN
JC		
JZ		
JO		
JS		
JP		

FLAGS	
Sign	
Zero	
Carry	
Overflow	
Parity	

Calculations

2. Update flags after every **CMP** instruction and calculate jumps

```
MOV ax,-12
CMP cx,12
JB L1
JNAE L2
JNGE L3
JL L4
```

	TAKEN	NOT TAKEN
JB		
JNAE		
JNGE		
JL		

Subtraction to Calculate Flags for unsigned numbers

2's Complement addition to calculate flags for signed number

3. Update register after every instruction

```
mov ax,014adh
mov bl,052q
xor al,0
xor al,bl
xor bl,al
xor bl,bl
```

Al	
Al	
Bl	
Bl	

COAL QUIZ 3 (B)

NAME: ____________________________ SECTION______ STUDENT ID: ____________

4. Convert the following **if condition** to equivalent Assembly code using conditional jumps

C Code	Assembly

`int al=129;`
`if(al>-5 && al<127)`
`{`
`al=255;`
`}`
`else`
`{`
`al=-128;`
`}`	

5. Convert the following Assembly code equivalent **C Code.**

Assembly	C Code

`mov ebx,10`
`mov ax,20`
`L1:`
`cmp ax,5`
`jbe L2`
`dec bx`
`sub ax,3`
`jmp L1`
`L2:`	

6. Dry run given C code and write final values of all integers.Convert the following **loops** to equivalent Assembly code using conditional jumps

C Code	Assembly

`int bl=-4,cl=0;`
`while(bl<15)`
`{`
`for(int al=8;al>-5;al=al-3)`
`{`
`cl++;`
`}`
`bl=bl+7;`
`}`	

Al	
Bl	
Cl	